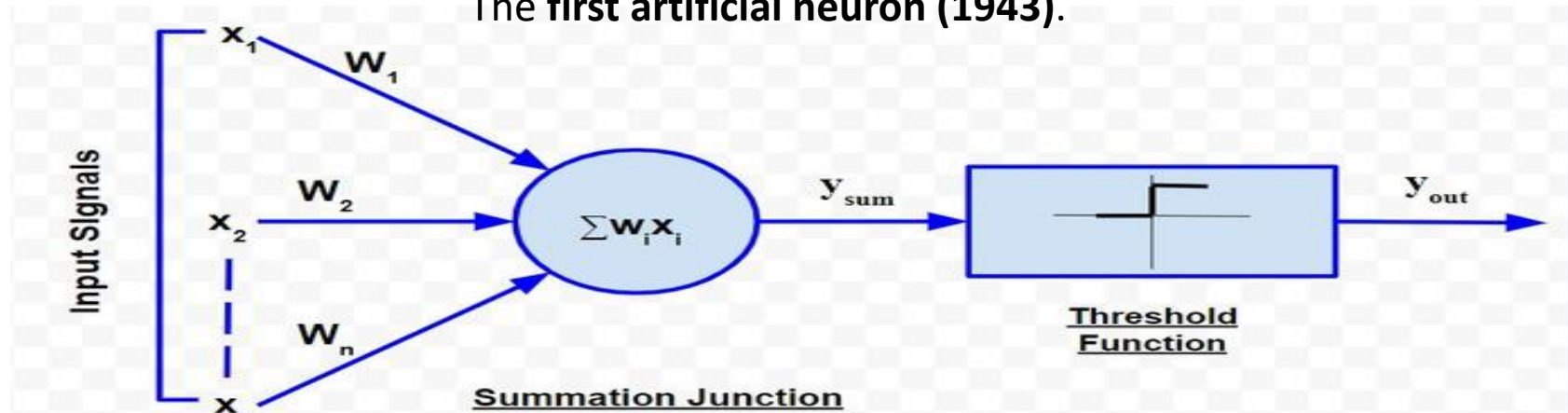


Activation Functions and Feed Forward Networks

McCulloch-Pitts model (Linear Threshold Gate)

The first artificial neuron (1943).



1. Binary Neuron:

- The model represents neurons as binary devices that output either 0 or 1 (binary output), based on the sum of their inputs.
- It does not account for more complex activations, making it a simpler model compared to modern artificial neurons.

2. Inputs and Weights:

- The neuron receives several **binary inputs** ($x_1, x_2, \dots, x_n$), each associated with a **weight** $w_1, w_2, \dots, w_n$.
- Each input represents either a signal being sent to the neuron or a lack of signal (binary 0 or 1).

3. Summation of Inputs:

- The weighted sum of the inputs is calculated:

$$S = w_1x_1 + w_2x_2 + \cdots + w_nx_n$$

- The sum represents the total "stimulus" to the neuron.

4. Threshold Function (Activation):

- The neuron has a **threshold** (θ), which determines whether the neuron fires (outputs 1) or not (outputs 0).
- If the weighted sum S exceeds the threshold θ , the output is 1. Otherwise, the output is 0.
- The output y is given by:

$$y = \begin{cases} 1 & \text{if } S \geq \theta \\ 0 & \text{if } S < \theta \end{cases}$$

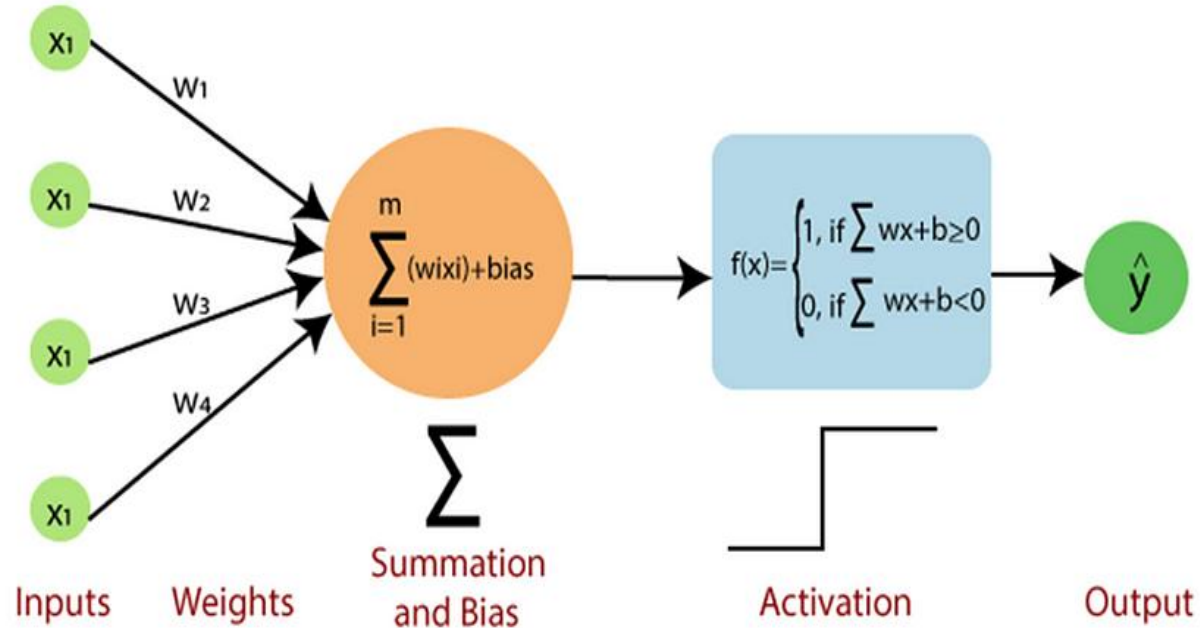
Unit Step Function:

 Boolean inputs	 Linear
 Boolean output	 Fixed Slope

$$y_{out} = \begin{cases} 1 & s \geq 0 \\ 0 & s < 0 \end{cases}$$

Single layer Perceptron

An improved MP neuron that can **learn weights automatically** from data.



Multi layer Perceptron

Drawback: The perceptron can only solve problems that are linearly separable (i.e., it can only classify data that can be separated by a straight line or hyperplane)

Comparison Table

Aspect	Single-Layer Perceptron (SLP)	Multi-Layer Perceptron (MLP)
Architecture	One input layer, one output layer (no hidden layers)	Input, hidden, and output layers
Problem Solvability	Solves only linearly separable problems (e.g., AND)	Solves both linear and non-linear problems (e.g., XOR)
Activation Function	Step function (binary output)	Non-linear functions (e.g., sigmoid, ReLU, tanh)
Learning Algorithm	Perceptron learning rule (no backpropagation)	Trained using backpropagation and gradient descent
Output	Binary (0 or 1)	Continuous (for regression) or multi-class (for classification)
Applications	Simple binary classification	Complex tasks (classification, regression, image recognition)
Complexity	Simple and limited	Complex and powerful

Implement AND function using McCulloch-Pitts Neuron

- Consider the truth table for AND function
- The M-P neuron has no particular training algorithm
- In M-Pneuron, only analysis is being performed.
- Hence, assume the weights be $w_1 = 1$ and $w_2 = 1$.

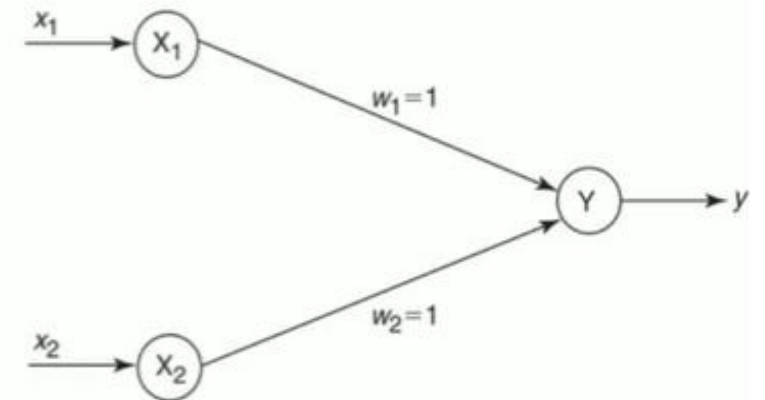
$$(1, 1), y_{in} = x_1 w_1 + x_2 w_2 = 1 \times 1 + 1 \times 1 = 2$$

$$(1, 0), y_{in} = x_1 w_1 + x_2 w_2 = 1 \times 1 + 0 \times 1 = 1$$

$$(0, 1), y_{in} = x_1 w_1 + x_2 w_2 = 0 \times 1 + 1 \times 1 = 1$$

$$(0, 0), y_{in} = x_1 w_1 + x_2 w_2 = 0 \times 1 + 0 \times 1 = 0$$

x_1	x_2	y
1	1	1
1	0	0
0	1	0
0	0	0



- Thus, the output of neuron Y can be written

$$y = f(y_{in}) = \begin{cases} 1 & \text{if } y_{in} \geq 2 \\ 0 & \text{if } y_{in} < 2 \end{cases}$$

2. Implement the following using M-P neuron

x_1	x_2	y
0	0	0
0	1	0
1	0	1
1	1	0

Assume: $w_1=1$, $w_2=1$

$$y_{in} = x_1 w_1 + x_2 w_2$$

$$(1, 1), y_{in} = 1 \times 1 + 1 \times 1 = 2$$

$$(1, 0), y_{in} = 1 \times 1 + 0 \times 1 = 1$$

$$(0, 1), y_{in} = 0 \times 1 + 1 \times 1 = 1$$

$$(0, 0), y_{in} = 0 \times 1 + 0 \times 1 = 0$$

- Hence, assume the weights be $w_1 = 1$ and $w_2 = -1$.

$$y_{in} = x_1 w_1 + x_2 w_2$$

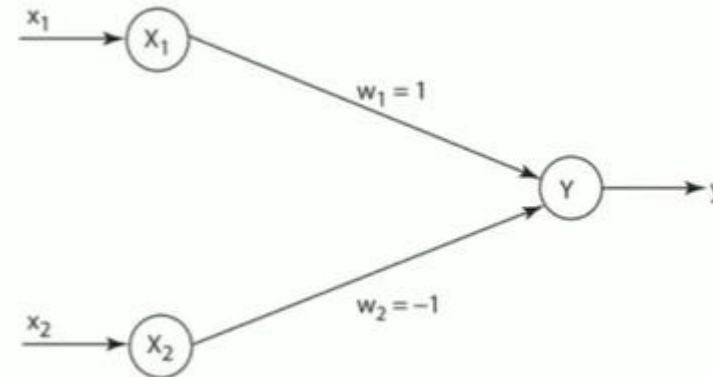
$$(1, 1), y_{in} = 1 \times 1 + 1 \times -1 = 0$$

$$(1, 0), y_{in} = 1 \times 1 + 0 \times -1 = 1$$

$$(0, 1), y_{in} = 0 \times 1 + 1 \times -1 = -1$$

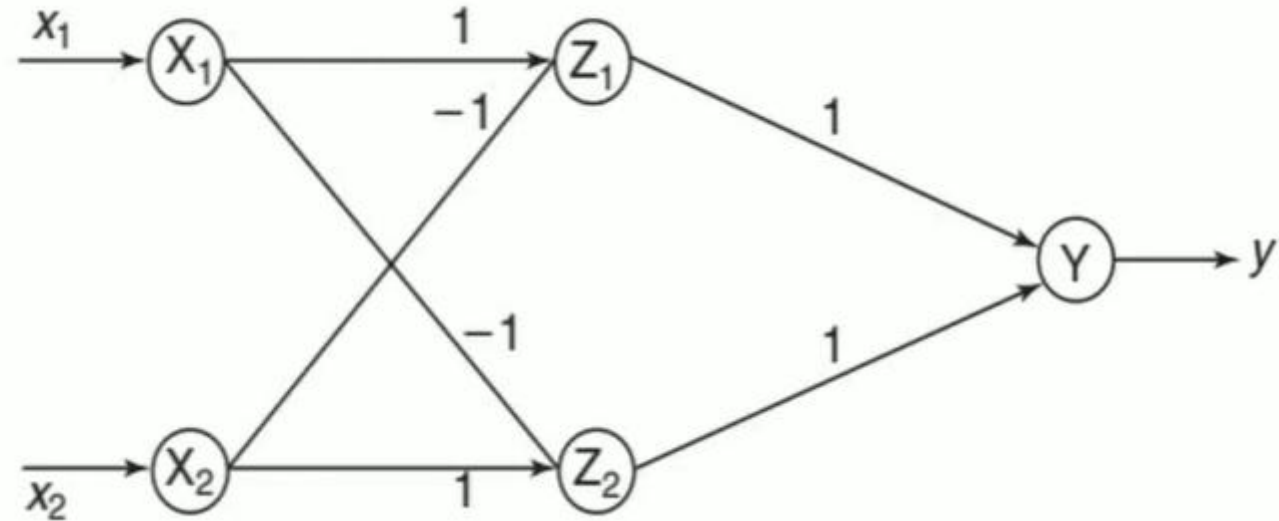
$$(0, 0), y_{in} = 0 \times 1 + 0 \times -1 = 0$$

From the calculated net inputs,
now it is possible to fire the
neuron for input (1, 0) only by
fixing a threshold of 1,
i.e., $\theta \geq 1$ for Y unit.



Implement XOR function using M-P neuron

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0



$$y = x_1 \overline{x_2} + \overline{x_1} x_2$$

$$y = z_1 + z_2$$

where

$$z_1 = x_1 \overline{x_2} \quad (\text{function 1})$$

$$z_2 = \overline{x_1} x_2 \quad (\text{function 2})$$

$$y = z_1 \text{ (OR) } z_2 \quad (\text{function 3})$$

- First function $z_1 = x_1 \overline{x_2}$
- The truth table for function z_1
- Assume the weights are initialized to

$$w_{11} = 1; \quad w_{21} = -1$$

- Calculate the net inputs,

$$(0, 0), z_{1in} = 0 \times 1 + 0 \times -1 = 0$$

$$(0, 1), z_{1in} = 0 \times 1 + 1 \times -1 = -1$$

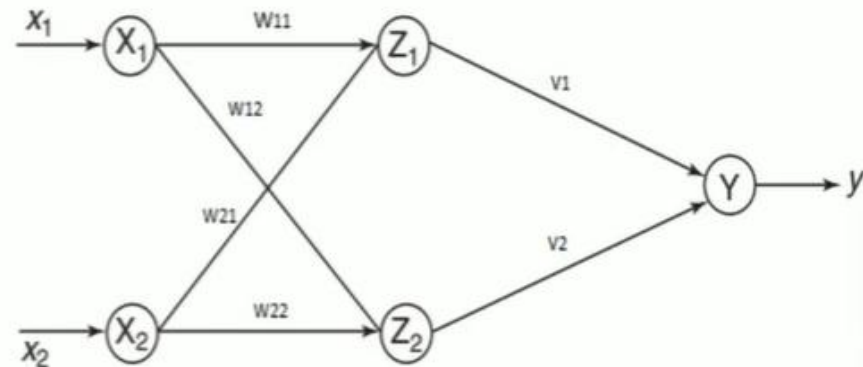
$$(1, 0), z_{1in} = 1 \times 1 + 0 \times -1 = 1$$

$$(1, 1), z_{1in} = 1 \times 1 + 1 \times -1 = 0$$

- If the $\theta=1$ then the neuron fires.
- Hence $w_{11} = 1; \quad w_{21} = -1$

$$f(y_{in}) = \begin{cases} 1 & \text{if } y_{in} \geq \theta \\ 0 & \text{if } y_{in} < \theta \end{cases}$$

x_1	x_2	z_1
0	0	0
0	1	0
1	0	1
1	1	0



- Third function $y = z_1$ (OR) z_2

- The truth table for function y

$$y_{in} = z_1 v_1 + z_2 v_2$$

- Assume the weights are initialized to

$$v_1 = v_2 = 1$$

- Calculate the net inputs, $f(y_{in}) = \begin{cases} 1 & \text{if } y_{in} \geq \theta \\ 0 & \text{if } y_{in} < \theta \end{cases}$

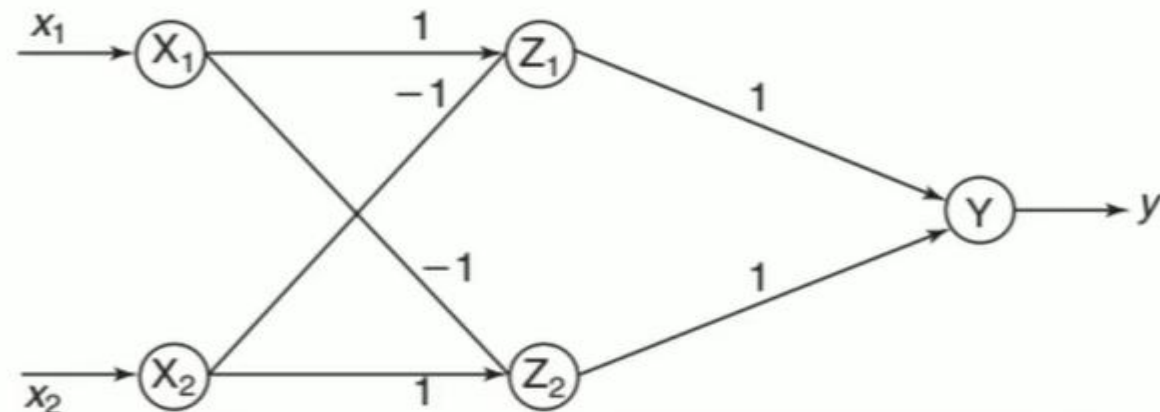
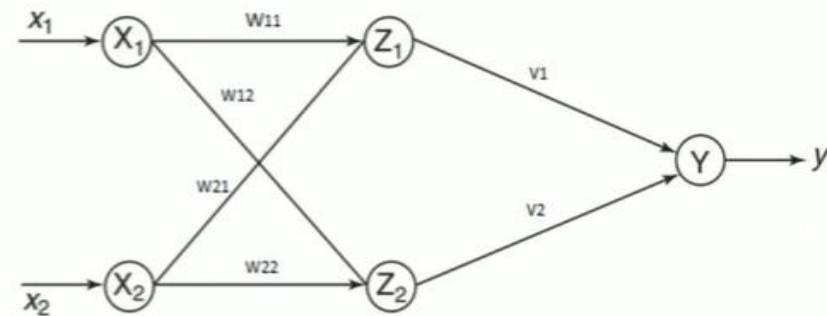
$$(0, 0), y_{in} = 0 \times 1 + 0 \times 1 = 0$$

$$(0, 1), y_{in} = 0 \times 1 + 1 \times 1 = 1$$

$$(1, 0), y_{in} = 1 \times 1 + 0 \times 1 = 1$$

$$(0, 0), y_{in} = 0 \times 1 + 0 \times 1 = 0$$

x_1	x_2	y	z_1	z_2
0	0	0	0	0
0	1	1	0	1
1	0	1	1	0
1	1	0	0	0



Activation function

- **An Activation Function** decides whether a neuron should be activated or not.
- This means that it will decide whether the neuron's input to the network is important or not in the process of prediction using simpler mathematical operations.
- Other name - *Transfer Function* in Artificial Neural Network.
- The primary role of the Activation Function is to **transform** the summed weighted input from the node into an output value to be fed to the next hidden layer or as output.

Date: 13-04-2026

Ttoo,
The HR Manager / Concerned Authority
Respected Organizations (Wipro, Infosys, TCS)

Subject: Request for Summer Internship Opportunity in IT Organizations

Dear Sir/Madam,

I am writing to formally request a summer internship opportunity at Respected organizations for Namepalli Lokeshwar Varma [CH.EN.U4CCE24020] pursuing B.Tech Electronics and Communication students from Amrita Vishwa Vidyapeetham, Chennai campus. As part of their academic curriculum, they are required to undergo an industrial internship to gain practical exposure and industry experience.

The students are applying to reputed IT organizations such as Wipro, Infosys, and TCS. An internship in these organizations will provide them with valuable hands-on experience and exposure to real-world engineering and technological practices.

We kindly request you to consider their applications and grant them an opportunity for a minimum internship duration of 30 days.

Thank you for your time and consideration.

Yours Sincerely,

(Dr. C.Ganesh Kumar)
CHAIRMAN
Department of Electronics and Communication
Amrita School of Engineering
Amrita Vishwa Vidyapeetham
Thiruvallur, Chennai – 601 103